

## **Final Report Draft**

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## Introduction

In this capstone, we wanted to answer one big question: “How do geography, demography and time shape cancer burden and general health in the United States?”. And we split into three research objectives, which are:

- Whether gender, broad region and survey year are associated with the risk level of invasive cancer incidence and mortality. & Investigates whether race / ethnicity is linked to high-versus low-risk colorectal-cancer mortality.
- Investigate whether cancer incidence rates in the United States differ significantly across states and ethnic groups, and whether there is an interaction effect between these two factors.
- Based on demographic and behavioral variables, the third analysis method employed a regression approach to model the probability of individuals falling into three self-reported health categories, “Very Healthy,” “Moderately Healthy,” or “Unhealthy”.

## Method

In this capstone we applied three core methods covered in this course. We first used the Chi-Square test of independence for the first question. Secondly, two-way ANOVA to evaluate whether mean invasive-cancer incidence differs across state and ethnic groups. Finally, they conducted multinomial logistic regression to predict HealthGroup from gender, detailed age bands and race/ethnicity dummies.

## Analysis

### Part1: Chi-Square Test

- Is there a significant association between gender, region, year and the risk level of invasive cancer incidence/mortality in the U.S?

For this analysis, we use the age-adjusted rate because it allows for fair comparison of cancer incidence and mortality across states, genders, and years, removing the influence of different age structures in populations. After filtering the dataset to include only "Invasive cancer (all sites combined)" with age-adjusted rates and known gender, the final sample included 160 records for incidence (82 female, 78 male). The age-adjusted incidence rate ranged from 361.7 to 560.1, with a median of 447.4.

```
incidence_df <- subset(
  data,
  Question == "Invasive cancer (all sites combined), incidence" &
  DataValueType == "Age-adjusted Rate" &
  Stratification1 %in% c("Male", "Female") &
  !is.na(DataValue)
)
```

Because of the lack significant differences among individual states, we regrouped states into broader regions (East vs. West) based on U.S. Census Bureau classifications to increase statistical power and facilitate interpretation.

```
west_states <- c("CA", "OR", "WA", "NV", "ID", "MT", "WY", "CO", "UT", "AZ", "NM", "AK", "HI")
mortality_df$Region <- ifelse(mortality_df$LocationAbbr %in% west_states, "West", "East")
```

To assess whether there is a statistically significant association between gender, region, and the risk level (high/low) of invasive cancer incidence and mortality, we categorized rates above the median as "High risk" and those below as "Low risk." We then used the Chi-square test of independence to compare the distributions.

```
# High/Low Risk by Median
median_mortality <- median(mortality_df$DataValue, na.rm = TRUE)
mortality_df$RiskLevel <- ifelse(mortality_df$DataValue >= median_mortality, "High", "Low")
```

## Incidence

### Gender vs. Risk Level:

The contingency table showed a much higher proportion of high-risk males (63) compared to females (17), while most females were classified as low risk (65). The Chi-square test result was highly significant ( $X^2 = 55.26$ ,  $p < 0.001$ ), indicating a strong association between gender and invasive cancer incidence risk level.

### Region vs. Risk Level:

Among eastern states, high-risk and low-risk cases were relatively balanced (73 vs 48), while in western states, low-risk cases dominated (32 vs 7). The Chi-square test was significant ( $X^2 = 19.53$ ,  $p < 0.001$ ), suggesting a regional difference in incidence risk level.

### Year vs. Risk Level:

No significant association was found between year (2015/2016) and risk level ( $X^2 = 0.03$ ,  $p = 0.87$ ).

## Mortality

### Gender vs. Risk Level:

The pattern was even stronger for mortality: nearly all high-risk cases were male (78 out of 81),

and almost all females were low risk. The Chi-square test was extremely significant ( $X^2 = 131.6$ ,  $p < 0.001$ ).

### Region vs. Risk Level:

Regional difference was not statistically significant for mortality ( $X^2 = 1.58$ ,  $p = 0.21$ ), meaning east-west region does not explain much of the mortality risk variation.

- Is there a significant association between race and the risk level (high vs. low) of age-adjusted mortality for colorectal cancer among U.S. adults?

We made a change on this question because the former topic has few data which contains less than 160 rows. To avoid reducing the credibility of the results, we switched to another type of cancer with more data.

A chi-square test of independence was used to examine the association between race and the risk level (high vs. low) of colorectal cancer mortality among U.S. people. The results revealed a significant association between race/ethnicity and mortality risk level ( $X^2 = 147.1$ ,  $df = 4$ ,  $p < 0.001$ ).

Specifically, most Black, non-Hispanic and White non-Hispanic individuals were classified as high-risk, and most Asian or Pacific Islander and Hispanic individuals were classified as low risk. American Indian or Alaska Native individuals were more evenly distributed between high- and low-risk groups. These findings suggest that substantial racial and ethnic disparities exist in colorectal cancer mortality risk in the United States.

```
data: table_race_risk_colorectal
x-squared = 147.1, df = 4, p-value < 2.2e-16
```

## Part2: ANOVA Analysis

- The Impact of State and Ethnicity on Cancer Incidence

We would like to investigate whether cancer incidence rates in the United States differ significantly across states and ethnic groups, and whether there is an interaction effect between these two factors. We use the same dataset from the CDC's U.S. Chronic Disease Indicators and focuses specifically on invasive cancer incidence across all sites combined(filtered), measured per 100,000 population.

We used 3 filters applied: we use Invasive cancer (all sites combined), incidence from Question; We select Per 100,000 for DataValueUnit and remove entries from DataValue where value is 0. Retained variables are DataValue, LocationDesc (state), Stratification1 (used as Ethnicity).

After successful transformation of the dataset, we can now build the regression model as below and then run the model in R to get a simulation:

$$\text{DataValue} = \beta_0 + \beta_1 * \text{LocationDesc} + \beta_2 * \text{Ethnicity} + \beta_3 * \text{LocationDesc} * \text{Ethnicity}$$

```

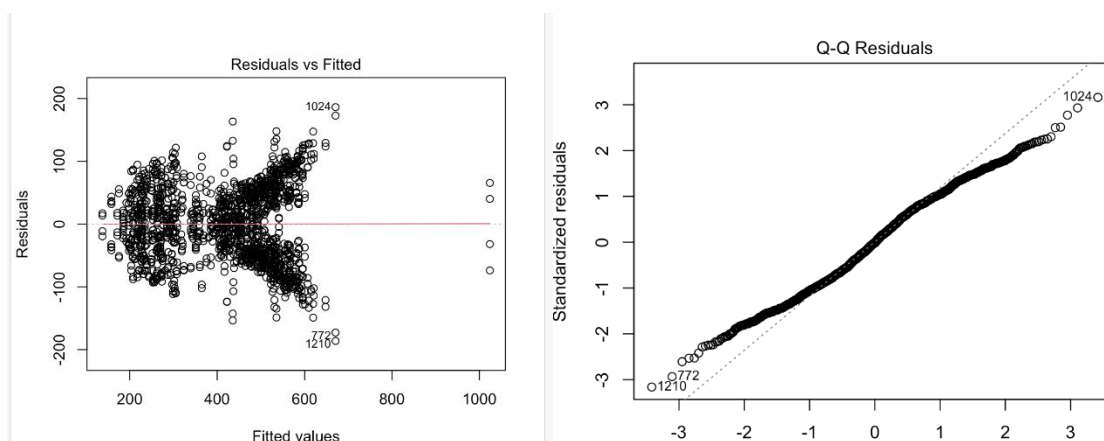
              Df Sum Sq Mean Sq F value Pr(>F)
LocationDesc   52  2171103   41752    9.031 <2e-16 ***
Ethnicity       7 17998105  2571158  556.150 <2e-16 ***
LocationDesc:Ethnicity 344  6782495   19717    4.265 <2e-16 ***
Residuals     1180  5455300    4623
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

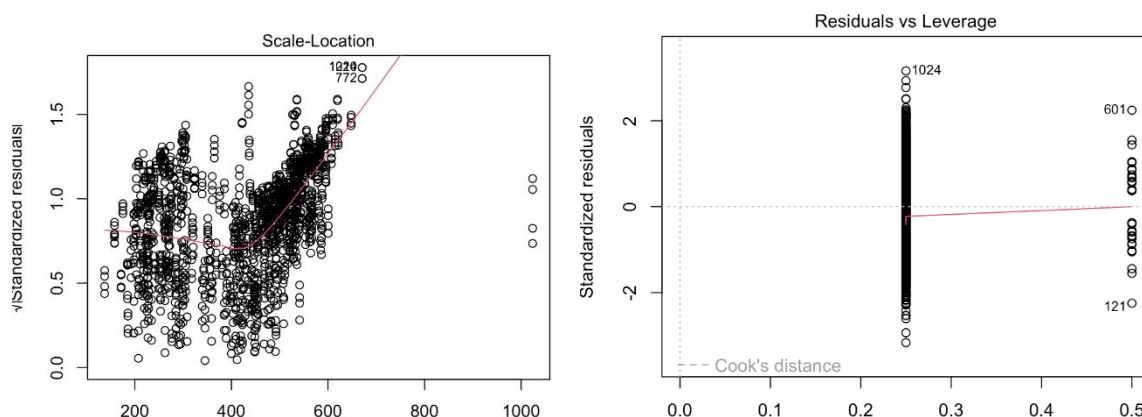
```

From the result table above, we can conclude that both State and Ethnicity have a significant effect on cancer incidence. This means that the average incidence rate of invasive cancer varies substantially between states and differs across ethnic groups. These differences are unlikely to be caused by random chance, but by those factors.

The interactive effect is also significant, suggesting the effect of ethnicity varies by state. What that means is that the pattern of cancer incidence among ethnic groups is not consistent across all states and certain ethnic groups may have higher or lower incidence rates depending on the specific state they reside in. This suggests that both geographic and demographic contexts jointly affect cancer risk, indicating the importance of localized public health strategies to improve cancer incidence rate.

Since we already have the model, we would need to do model Diagnostics to evaluate the model. There are 4 methods applied to this model: **Residuals vs Fitted**, **Q-Q Plot**, **Scale-Location Plot**, and **Residuals vs Leverage**.





From residuals & fitted graph, it shows a "funnel" shape indicating heteroscedasticity (non-constant variance). Q-Q plot shows Residuals mostly follow the normal line, with slight tail deviations indicating approximate normality with a few outliers. From scale-location plot, we can see trend line rises with fitted values, further confirming heteroscedasticity. The last plot – residuals and leverage, observations 1024, 601, and 121 exhibit high leverage and may be influential.

To reduce heteroscedasticity, applying a log transformation to DataValue can make residuals more uniform and improve model assumptions. Use robust variance estimation methods (e.g., sandwich estimators). These techniques provide more reliable standard errors and p-values when the assumptions of homoscedasticity and normality are violated.

### Part3: Logistics Regression

- What is the probability that an individual falls into “Very Healthy / Moderately Healthy / Unhealthy

Data Preparation (cleansing, preprocessing, feature engineering, feature selection)

As part of our preparatory work for the logistic regression analysis of the U.S. Chronic Disease Indicators dataset, we first assembled the necessary packages. After reading the CSV file, we first began data manipulation.

To narrow our focus, we ranked the frequency of each Topic and identified the six most prevalent topics. Among them (you can see the details in [Figure 1](#)), we chose **Health Status** as our research subject for the logistic regression model.

Filtering the original dataset by health status topic, and then we profiled each indicator by different questions versus data value units. From the summarized table, we easily notice that each question

is asked with the same frequency, the distribution is shown in [Figure 2](#). Based on the inspection, despite the various combinations of unit and value type, there are only five unique survey items (“questions”) in our Health Status subset.

To visualize these patterns, we produced faceted histograms for five representative health-status questions. The initial plots highlighted slight right-skewness and a small number of extreme values in certain indicators. We then quantified missingness in this subset in [Figure 3](#), and finally made a decision to remove those records with missing data values.

To verify the consistency of distribution for each variable, we overlaid the “before” and “after” histograms to confirm that our data cleaning only had minimal impact on the overall shape of each distribution. The before and after histograms are displayed in [Figure 4](#).

Finally, with a clean and well-understood dataset and clear distributional insights, we are now ready to proceed to the logistic regression modeling phase.

We have chosen to work with the crude prevalence measures (the percentage-based metrics) rather than the age-adjusted values, because age will be incorporated explicitly as a covariate in our regression models. By focusing on indicators all expressed on the same scale (percentages), we ensure that every variable is directly comparable, greatly simplifying both interpretation and analysis.

- Multinomial Logistic Regression Model

The logistic modeling part began by categorizing each Health Status observation. In our case, the percent-based “Crude Prevalence” values are classified into three ordered groups: “Very Healthy”, “Moderately Healthy”, and “Unhealthy”. We identified them by the first and third quantiles.

After that, we used `fastDummies` to one-hot encode each remaining demographic factor—gender, age, and race/ethnicity categories. To avoid multicollinearity, we dropped the first dummy of each set. Moreover, we fitted a multinomial logistic regression model, treating 'Unhealthy' as the reference level and predicting 'HealthGroup' from the ten binary predictors. [Figure 5](#) below summarizes the coefficients, their standard errors, z-values, and p-values for each non-reference outcome.

- Analysis

To gain valuable insights from the model summary, we focus on the significant odds ratios ( $p < 0.05$ ) from the multinomial logistic regression, interpret them relative to the omitted reference



group (Unhealthy, treated as the baseline), and highlight the most actionable findings. The odds ratios data are shown in Figure 6, and the following part will analyze them specifically

### **Age:**

So far, age is the strongest driver. We can see that for ages from 18 to 44, the odds ratio is up to 3150 for ModeratelyHealthy and 1056 times higher odds of VeryHealthy versus the baseline. As for 45-64, there are about 10 to 20 times higher odds in both classifications. These outcomes show that adults aged 18-40 have dramatically higher odds for ModeratelyHealthy and VeryHealthy, compared to the reference group (aged greater than 65). In this case, this younger group is thought of as a healthy status group based on our survey questions.

By designing and marketing wellness programs specifically for 18–44-year-olds, we can capitalize on their existing positive health behaviors and mindsets, helping cement healthy habits during the life stage when they are most receptive. At the same time, it reduces the incidence of chronic disease and healthcare costs down the road.

### **Gender:**

From our table, we found that women are 34.7 times more likely than the baseline group to fall into the ModeratelyHealthy bracket (few chronic issues/distress) and 7.1 times more likely to be in the VeryHealthy bracket. Whereas, men rarely land in ModeratelyHealthy ( $OR \approx 0$ ), but when they are healthy, they jump straight into the VeryHealthy class, 182.7 times more likely than baseline.

The severe polarization reminds us of the need for initial intervention for men. It is necessary to avoid and catch chronic disease early, before it becomes severe. Breaking through the situation of all-or-nothing in their health profile by tracking healthy metrics in daily life. As for women, their situation is not that bad, but they also have the space to improve, upgrading their health from moderate to a very healthy level.

### **Race/Ethnicity:**

The multinomial logistic regression results reveal a clear stratification of health status by race and ethnicity, using “Unhealthy” as the reference outcome. Four groups: White non-Hispanic, Hawaiian/Pacific Islander non-Hispanic, Asian non-Hispanic, and Multiracial non-Hispanic, showing consistently elevated odds of both moderate and very healthy status.

Specifically, White non-Hispanics are approximately 283 times more likely to report moderate health and 3,348 times more likely to report very healthy status compared with the reference group; Hawaiian/Pacific Islanders are 992 times and 394 times more likely in those same categories. In contrast, Hispanics do not differ significantly from the reference group in the moderate-health category, yet they exhibit a 37.9-fold increase in odds of very healthy status.

Based on the foregoing analysis, the next step we should take is to conduct in-depth qualitative interviews to uncover the community-driven practices that underpin their superior health outcomes, such as daily dietary patterns, social support networks, and routine preventive-care behaviors. Then, translate those insights into health promotion strategies that can be replicated in other populations.

Furthermore, to evaluate the predictive capacity of our model, the classification was assessed via a confusion matrix and visualized with a heatmap of the confusion table. We displayed it in Figure 7. Based on that, we created a flextable that aligns with several evaluated metrics, including recall, precision, specificity, and F1 score. This is shown in the following table:

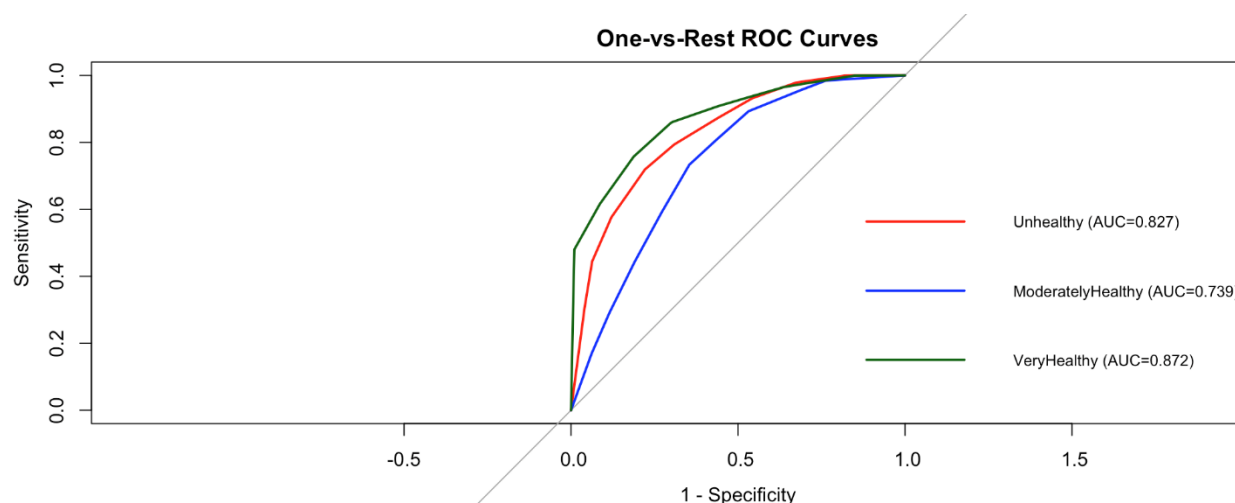
| <b>Class</b>             | <b>Recall</b> | <b>Precision</b> | <b>Specificity</b> | <b>F1</b> |
|--------------------------|---------------|------------------|--------------------|-----------|
| Class: Unhealthy         | 0.577         | 0.635            | 0.879              | 0.604     |
| Class: ModeratelyHealthy | 0.819         | 0.619            | 0.549              | 0.705     |
| Class: VeryHealthy       | 0.480         | 0.944            | 0.990              | 0.637     |

From this table, we found that our classifier is doing quite well in the middle. For Unhealthy, Specificity 0.879 means you rarely mis-label healthy people as unhealthy. Recall 0.577 means you correctly catch about 58% of truly unhealthy cases. Precision 0.635 shows that when you predict “Unhealthy,” you’re right about 64% of the time.

For ModeratelyHealthy, with Recall 0.819, means the model catches over 80% of that class, but Precision 0.619 means about 38% of your “ModeratelyHealthy” calls are false positives.

Regarding VeryHealthy, Recall is only 0.480, which represents the model missing more than half of truly very-healthy folks. However, Precision 0.944 means almost every “VeryHealthy” prediction is correct. Specificity 0.990 confirms almost nobody gets mislabeled “VeryHealthy.”

With the help of the ROC curve visualization, we concluded how well the model separates each health category from the others



**VeryHealthy (green, AUC  $\approx 0.872$ ):** This curve is located in the top-left corner more tightly than the others, meaning the model most reliably pulls out truly “very healthy” cases with both high sensitivity and specificity.

**Unhealthy (red, AUC  $\approx 0.827$ ):** With an AUC just over 0.82, the model also does a strong job of flagging “unhealthy” individuals while avoiding false alarms.

**ModeratelyHealthy (blue, AUC  $\approx 0.739$ ):** The lowest AUC indicates that “moderately healthy” is the hardest class to pin down. The curve is closer to the diagonal, so you must trade off sensitivity versus specificity more dramatically.

## Interpretation & Conclusions

From Chi-Square test analysis, we found that there is a strong association between gender and **invasive-cancer** risk: 63 of 80 male observations fell into the high-risk cell, whereas females were mostly low risk. Eastern and Western states also differed, but the survey year had no effect. For mortality, the gender gap widened further while the regional gap disappeared. **Colorectal cancer** mortality risk is not evenly distributed across races. Black and White non-Hispanics were mostly high-risk, whereas Asian/Pacific Islander and Hispanic populations were largely low risk. These findings point to persistent demographic inequities that general awareness campaigns alone are unlikely to fix.

Moreover, the ANOVA results provide strong statistical evidence that both geographic location (State) and demographic characteristics (Ethnicity) significantly influence cancer incidence rates in the United States. Furthermore, the significant interaction effect between State and Ethnicity indicates that the relationship between ethnicity and cancer risk is not uniform across different

states. These findings highlight the importance of considering both location-specific and population-specific factors when designing and implementing public health interventions.

However, diagnostic plots reveal signs of heteroscedasticity and the presence of influential observations, suggesting some violations of regression assumptions. To address these issues, a log transformation of the response variable and the application of robust variance estimation methods are recommended. These adjustments will help ensure the reliability of the statistical inference and improve the model's explanatory power. Ultimately, the results support a tailored approach to cancer prevention efforts that considers both demographic diversity and local health contexts.

Regarding the third method, our multinomial logistic model shows that demographic factors: age, gender, and race/ethnicity, strongly predict actual health status (measured by chronic conditions, self-rated poor health, and frequent distress).

Younger adults (18–44) overwhelm both the “Moderately Healthy” and “Very Healthy” categories, with odds more than a thousand times higher than the 65+ reference group.

Women are far more likely than the baseline group to fall into “Moderately Healthy,” whereas men exhibit a true “all-or-nothing” pattern, rarely landing in the middle but showing very high odds of being “Very Healthy” when they are healthy.

Among race/ethnicity, White non-Hispanics, Hawaiian/Pacific Islanders, Asians, and Multiracial respondents consistently enjoy higher odds of better health, while Hispanics only stand out in the “Very Healthy” class.

If we institute a quarterly “Health Status Tracking” survey, we can assess the real-world impact of our interventions promptly and iteratively refine our outreach strategies based on empirical evidence by continuously tracking how quickly each group moves between these three health tiers.

## Appendix

|   | Topic                                           | n     | Pct  |
|---|-------------------------------------------------|-------|------|
| 1 | Cardiovascular Disease                          | 30709 | 9.93 |
| 2 | Chronic Obstructive Pulmonary Disease           | 26951 | 8.72 |
| 3 | Nutrition, Physical Activity, and Weight Status | 26069 | 8.43 |
| 4 | Health Status                                   | 25612 | 8.28 |
| 5 | Alcohol                                         | 25321 | 8.19 |
| 6 | Cancer                                          | 22358 | 7.23 |

Figure 1 Top 6 Chronic Diseases

|    | DataValueUnit | Question                                              | DataValueType           | n    |
|----|---------------|-------------------------------------------------------|-------------------------|------|
| 1  | %             | 2 or more chronic conditions among adults             | Crude Prevalence        | 2860 |
| 2  | %             | Fair or poor self-rated health status among adults    | Crude Prevalence        | 2860 |
| 3  | %             | Frequent physical distress among adults               | Crude Prevalence        | 2860 |
| 4  | Number        | Average recent physically unhealthy days among adults | Crude Mean              | 2860 |
| 5  | Number        | Recent activity limitation among adults               | Crude Mean              | 2860 |
| 6  | %             | 2 or more chronic conditions among adults             | Age-adjusted Prevalence | 2200 |
| 7  | %             | Fair or poor self-rated health status among adults    | Age-adjusted Prevalence | 2200 |
| 8  | %             | Frequent physical distress among adults               | Age-adjusted Prevalence | 2200 |
| 9  | Number        | Average recent physically unhealthy days among adults | Age-adjusted Mean       | 2200 |
| 10 | Number        | Recent activity limitation among adults               | Age-adjusted Mean       | 2200 |
| 11 | Years         | Life expectancy at birth                              | Number                  | 312  |

Figure 2 Health Status Question Distribution

```
# A tibble: 5 × 2
```

| Question                                                | pct_missing |
|---------------------------------------------------------|-------------|
| <chr>                                                   | <dbl>       |
| 1 2 or more chronic conditions among adults             | 25.8        |
| 2 Average recent physically unhealthy days among adults | 22.6        |
| 3 Fair or poor self-rated health status among adults    | 25.0        |
| 4 Frequent physical distress among adults               | 30.1        |
| 5 Recent activity limitation among adults               | 24.2        |

Figure 3 Missing Value Percentage for Each Question

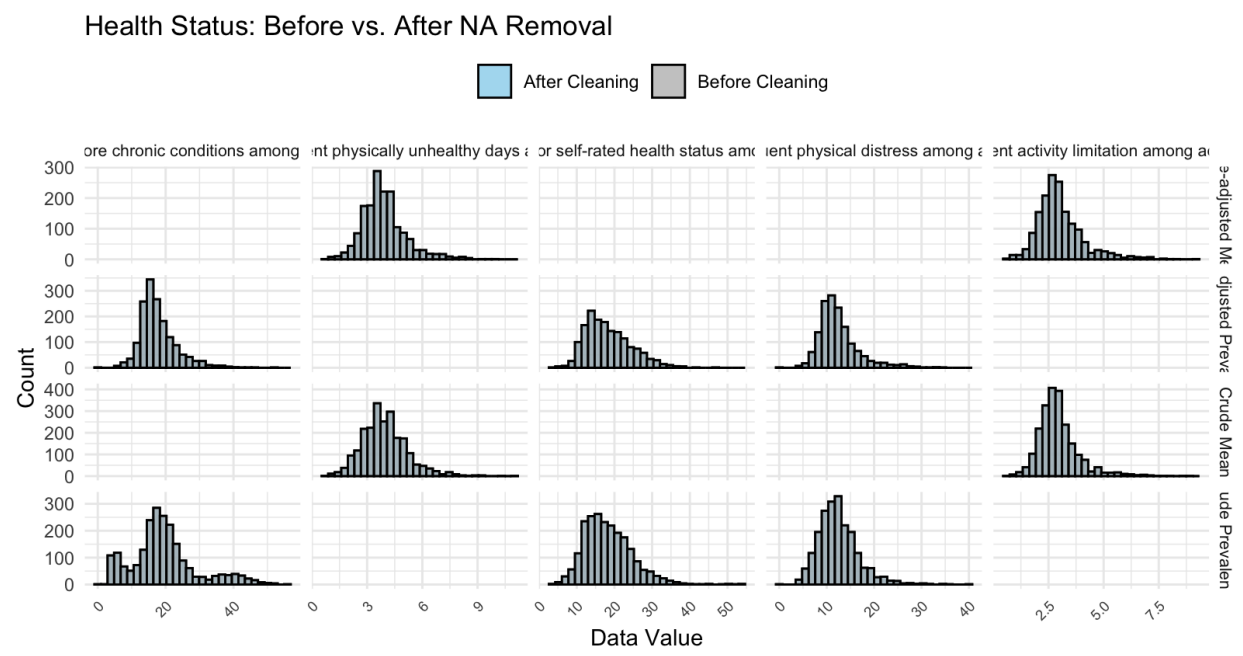
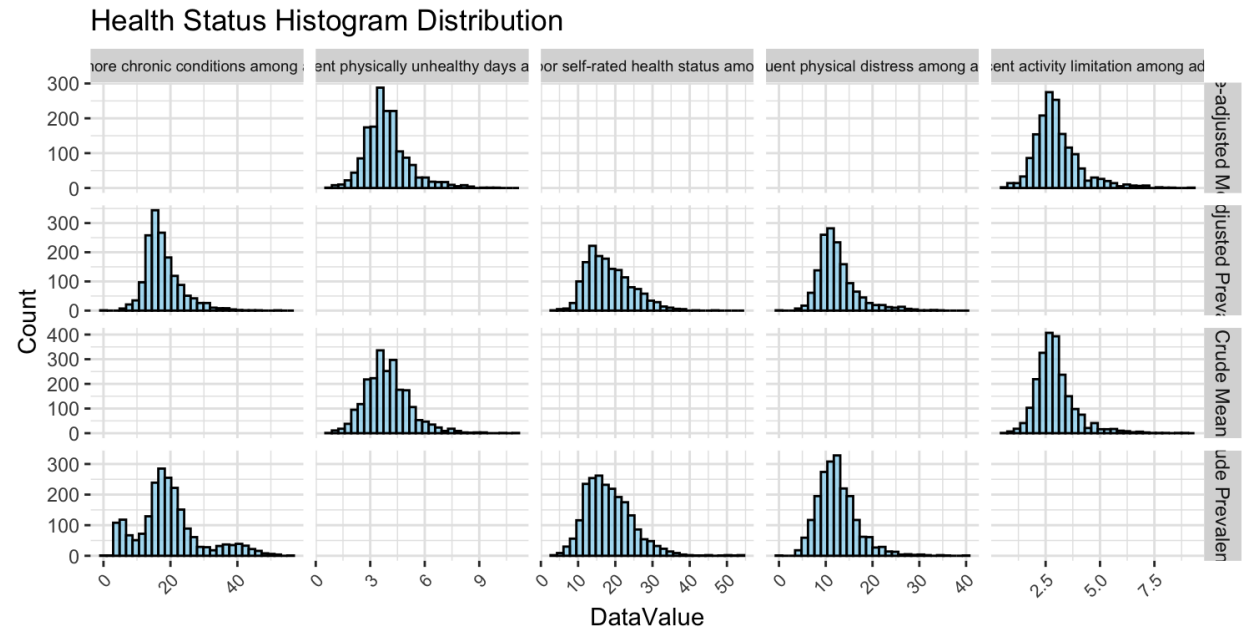


Figure 4 Before and After Histograms

| Outcome           | Predictor                                      | Coefficient | StdError | Zvalue   | Pvalue |
|-------------------|------------------------------------------------|-------------|----------|----------|--------|
| ModeratelyHealthy | (Intercept)                                    | -1.0174     | 0.0892   | -11.4021 | 0.0000 |
| ModeratelyHealthy | Male                                           | -6.1596     | 1.0013   | -6.1515  | 0.0000 |
| ModeratelyHealthy | Female                                         | 3.5456      | 0.2042   | 17.3619  | 0.0000 |
| ModeratelyHealthy | Age_18_44                                      | 8.0552      | 1.0191   | 7.9041   | 0.0000 |
| ModeratelyHealthy | Age_45_64                                      | 2.3843      | 0.1367   | 17.4414  | 0.0000 |
| ModeratelyHealthy | White__non_Hispanic                            | 5.6460      | 1.0127   | 5.5754   | 0.0000 |
| ModeratelyHealthy | Black__non_Hispanic                            | 10.8975     | 23.6263  | 0.4612   | 0.6446 |
| ModeratelyHealthy | Hispanic                                       | 18.8928     | 23.6467  | 0.7990   | 0.4243 |
| ModeratelyHealthy | Asian__non_Hispanic                            | 1.5899      | 0.1228   | 12.9508  | 0.0000 |
| ModeratelyHealthy | American_Indian_or_Alaska_Native__non_Hispanic | 4.2924      | 1.0181   | 4.2159   | 0.0000 |
| ModeratelyHealthy | Multiracial__non_Hispanic                      | 2.7371      | 0.1560   | 17.5451  | 0.0000 |
| ModeratelyHealthy | Hawaiian_or_Pacific_Islander__non_Hispanic     | 6.9002      | 1.0115   | 6.8217   | 0.0000 |
| VeryHealthy       | (Intercept)                                    | 2.0589      | 0.1408   | 14.6207  | 0.0000 |
| VeryHealthy       | Male                                           | 5.2076      | 1.0169   | 5.1209   | 0.0000 |
| VeryHealthy       | Female                                         | 1.9575      | 0.1525   | 12.8401  | 0.0000 |
| VeryHealthy       | Age_18_44                                      | 6.9619      | 1.0092   | 6.8982   | 0.0000 |
| VeryHealthy       | Age_45_64                                      | 2.9631      | 1.0727   | 2.7624   | 0.0057 |
| VeryHealthy       | White__non_Hispanic                            | 10.8956     | 1.4182   | 7.6829   | 0.0000 |
| VeryHealthy       | Black__non_Hispanic                            | 0.0175      | 0.1597   | 0.1098   | 0.9126 |
| VeryHealthy       | Hispanic                                       | 3.6362      | 1.0326   | 3.5215   | 0.0004 |
| VeryHealthy       | Asian__non_Hispanic                            | 1.0076      | 0.1335   | 7.5489   | 0.0000 |
| VeryHealthy       | American_Indian_or_Alaska_Native__non_Hispanic | 4.5792      | 1.0156   | 4.5090   | 0.0000 |

Figure 5 Non-Reference Outcomes

| Outcome           | Predictor                                      | Odds Ratio (p < 0.05) |
|-------------------|------------------------------------------------|-----------------------|
| ModeratelyHealthy | (Intercept)                                    | 0.36                  |
| ModeratelyHealthy | Male                                           | 0.00                  |
| ModeratelyHealthy | Female                                         | 34.66                 |
| ModeratelyHealthy | Age_18_44                                      | 3,150.00              |
| ModeratelyHealthy | Age_45_64                                      | 10.85                 |
| ModeratelyHealthy | White__non_Hispanic                            | 283.16                |
| ModeratelyHealthy | Asian__non_Hispanic                            | 4.90                  |
| ModeratelyHealthy | American_Indian_or_Alaska_Native__non_Hispanic | 73.14                 |
| ModeratelyHealthy | Multiracial__non_Hispanic                      | 15.44                 |
| ModeratelyHealthy | Hawaiian_or_Pacific_Islander__non_Hispanic     | 992.48                |
| VeryHealthy       | (Intercept)                                    | 7.84                  |
| VeryHealthy       | Male                                           | 182.66                |
| VeryHealthy       | Female                                         | 7.08                  |
| VeryHealthy       | Age_18_44                                      | 1,055.69              |
| VeryHealthy       | Age_45_64                                      | 19.36                 |
| VeryHealthy       | White__non_Hispanic                            | 53,937.19             |
| VeryHealthy       | Hispanic                                       | 37.95                 |
| VeryHealthy       | Asian__non_Hispanic                            | 2.74                  |
| VeryHealthy       | American_Indian_or_Alaska_Native__non_Hispanic | 97.43                 |
| VeryHealthy       | Multiracial__non_Hispanic                      | 9.68                  |
| VeryHealthy       | Hawaiian_or_Pacific_Islander__non_Hispanic     | 394.36                |

Figure 6 Odds Ratio against Baseline

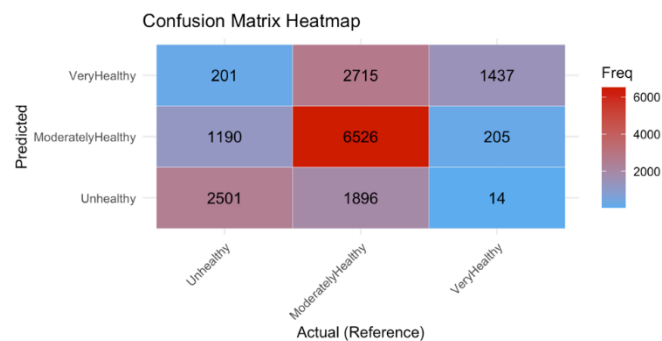


Figure 7 Confusion Matrix



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